

HOUSE PRICE PREDICTION – PROJECT SUMMARY

Introduction

The objective of this project was to build a machine learning model capable of predicting house prices based on various property features. The dataset contained information about housing characteristics such as area, number of bedrooms, bathrooms, stories, parking availability, furnishing status, and other amenities.

Data Exploration and Preprocessing

The dataset was loaded and explored using Pandas. The dataset consisted of 545 records and 13 features. Missing values and duplicate records were checked, and no significant missing data was found. Categorical variables such as main road access, guest room availability, basement, air conditioning, and furnishing status were converted into numerical values to make them suitable for machine learning algorithms.

Model Building and Evaluation

Two regression models were developed and compared:

1. Linear Regression
2. Random Forest Regressor

The dataset was divided into training and testing sets using an 80:20 ratio. The performance of the models was evaluated using MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), and R^2 Score.

The Random Forest Regressor achieved an R^2 score of approximately 0.61, indicating that the model could explain around 61% of the variation in house prices and performed better than Linear Regression.

Key Findings

- House area was one of the most influential factors affecting house prices.
- Properties with more bathrooms, additional stories, and air conditioning generally had higher prices.
- Houses located in preferred areas were significantly more expensive.
- The Random Forest model provided better prediction accuracy than the Linear Regression model.

Conclusion

This project demonstrated that machine learning techniques can effectively predict house prices based on property characteristics. Real estate businesses can use such predictive models to estimate fair market prices and make better investment decisions. The findings suggest that premium amenities and desirable locations play a significant role in determining housing prices.